



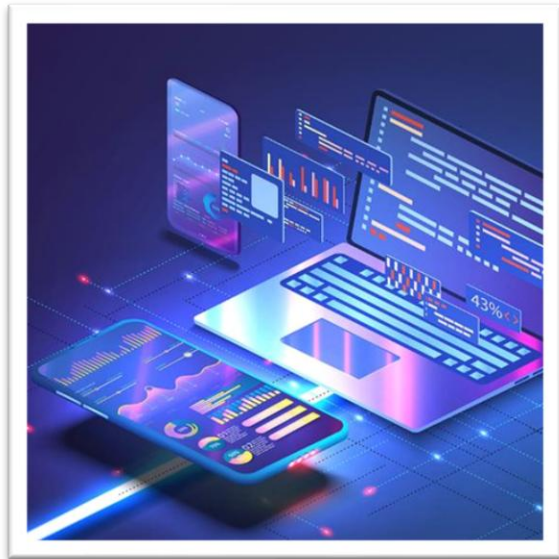
SwadeshAI: Building India's AI Sovereignty through Fundamental Research

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XXI International Conference on Public Policy
& Management
02/09/2026

What is necessary for sovereignty?

Let's look at established technologies.



Software



Cloud



Space tech

Success Story: ISRO

India became atmanirbhar in space *because* ISRO built indigenous space technology under the unique constraints and demands of a developing nation.

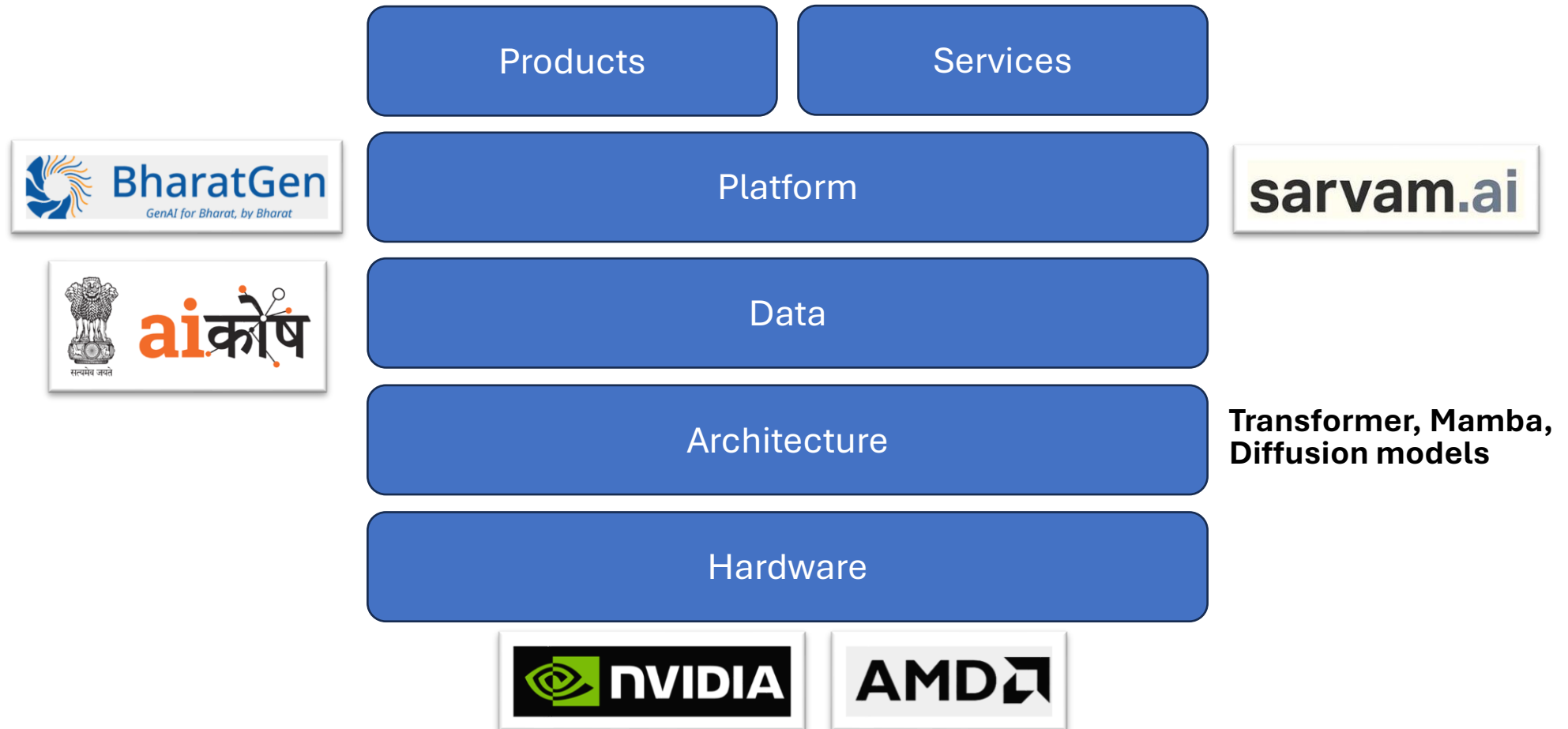
Major player in international commercial market.

- 10x revenue growth in last 3 yrs (INR 322 Cr -> 3246 Cr)¹
- INR 4368 Cr in Forex earnings in last 10 yrs²

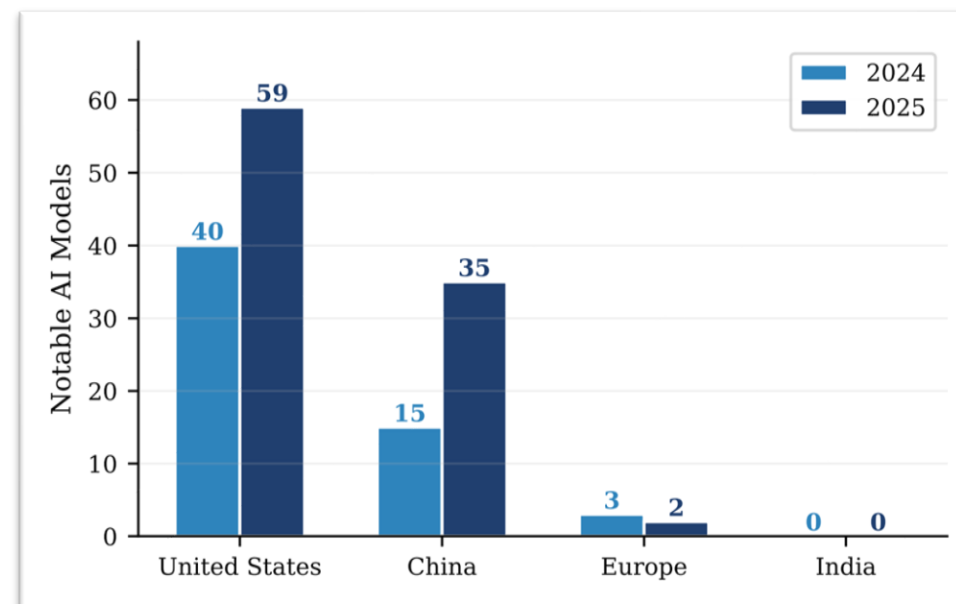
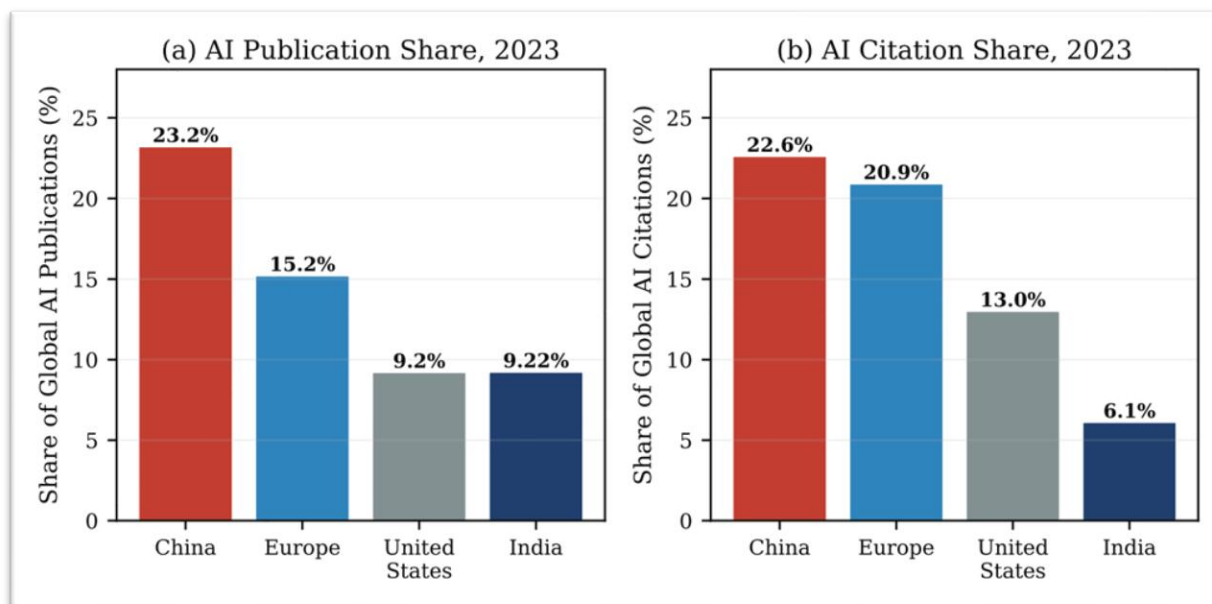
1. [Akashvani News](#), March 2025
2. [PIB Press Release](#), June 2026



Where do we stand in AI?

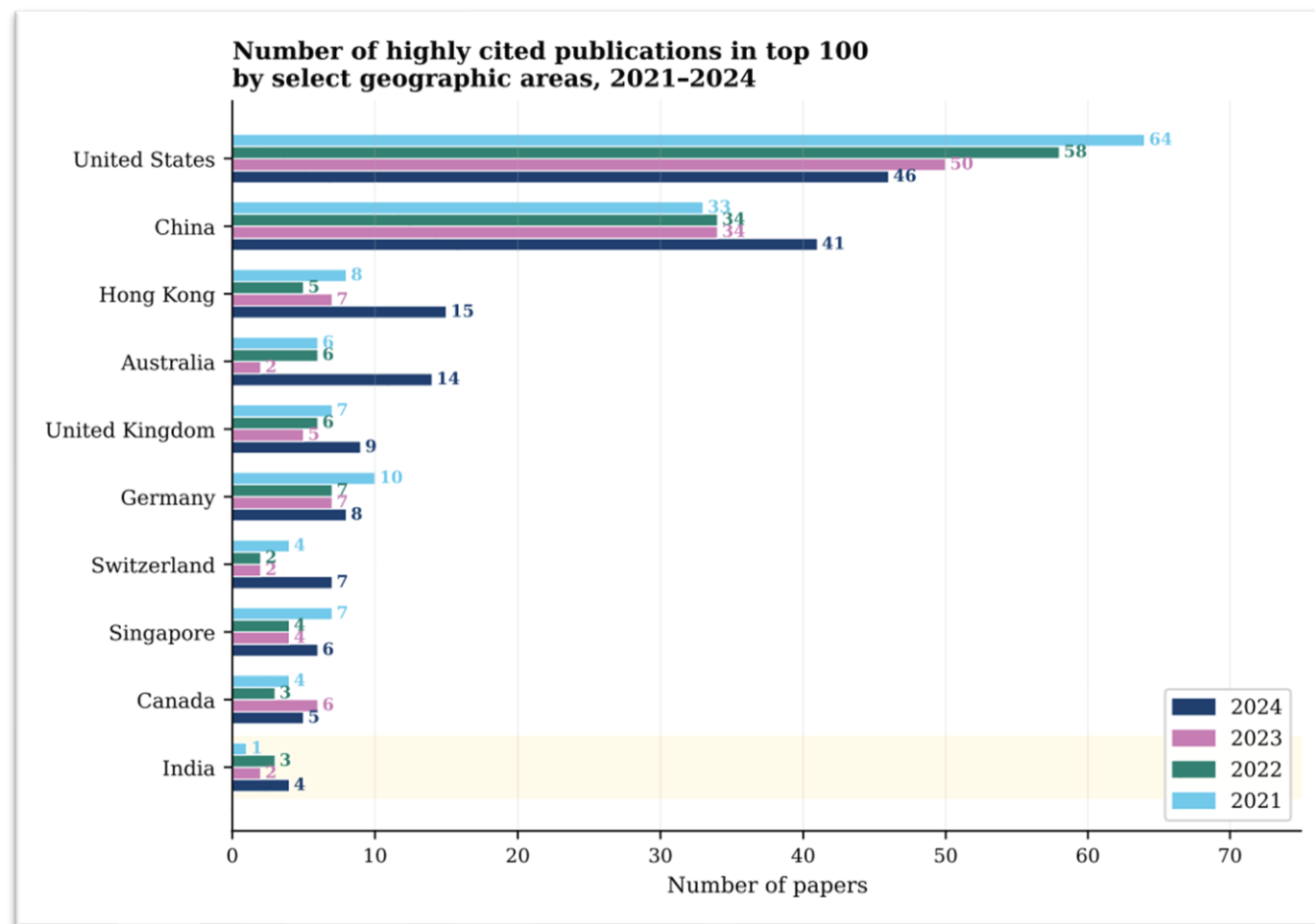


India has scale but no frontier impact



Sources: Stanford HAI AI Index 2026; Stanford HAI AI Index 2025; IndiaML / Change Engine trackers.

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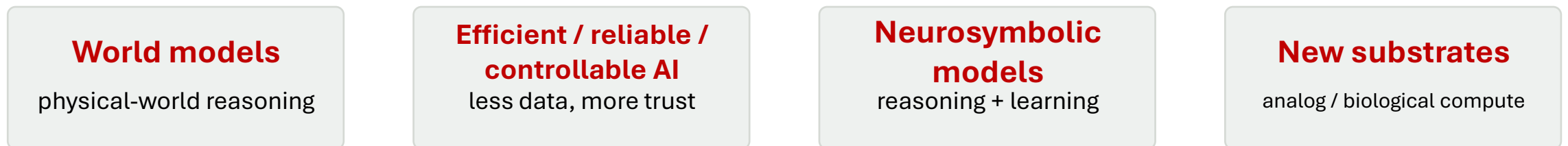


Sources: Stanford HAI AI Index 2026; Stanford HAI AI Index 2025; IndiaML / Change Engine trackers.

The stack changes. Sovereignty must survive the change.



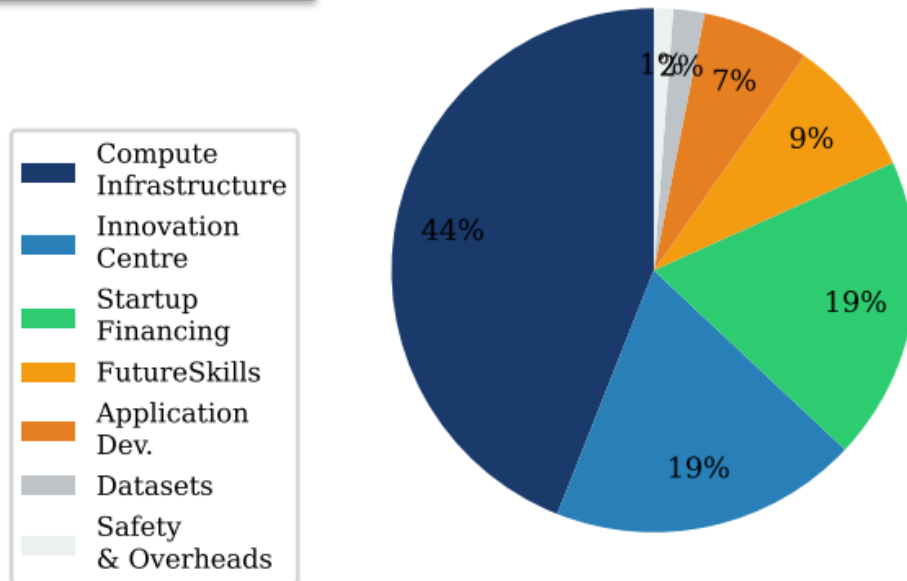
The next wave



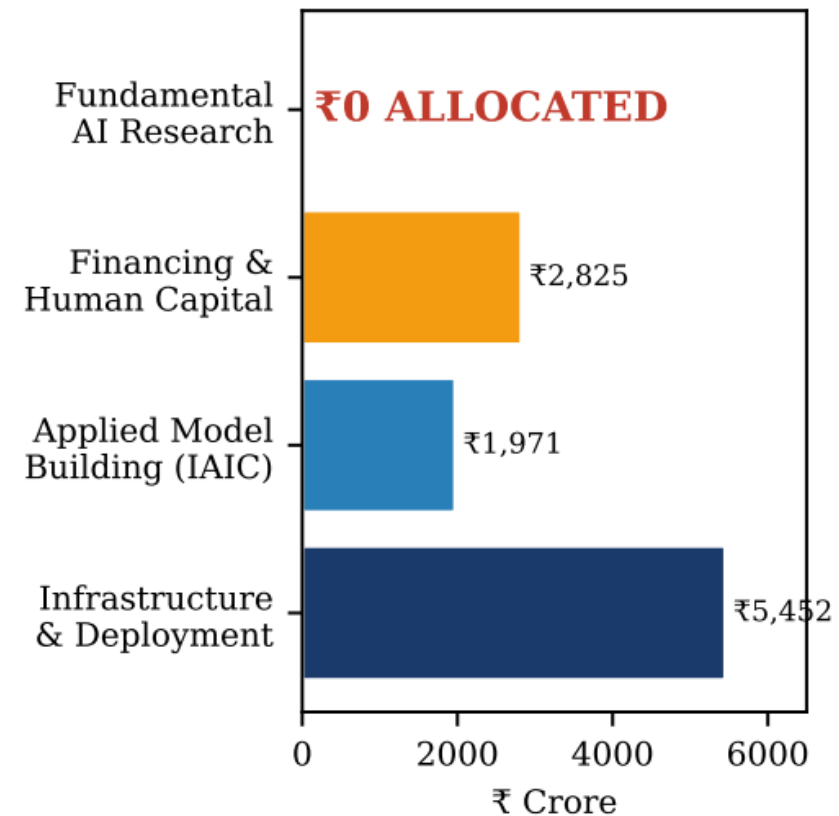
Diagnosis #1: application-first public investment



(a) Budget by Pillar
(Total: ₹10,372 Cr / \$1.25B)



(b) Research vs. Non-Research Allocation



Source: paper Figure 4; Parliamentary response Dec 2024 and Cabinet approval.

Diagnosis #1: application-first public investment

The writing is on the wall.

THE KEN

SLOW AND UNSTEADY

Rs 10K crore, 38K GPUs, and 12 firms later, India's AI mission is still looking for users



PIM

AI FEATURES

IndiaAI, Where are Your Research Papers?

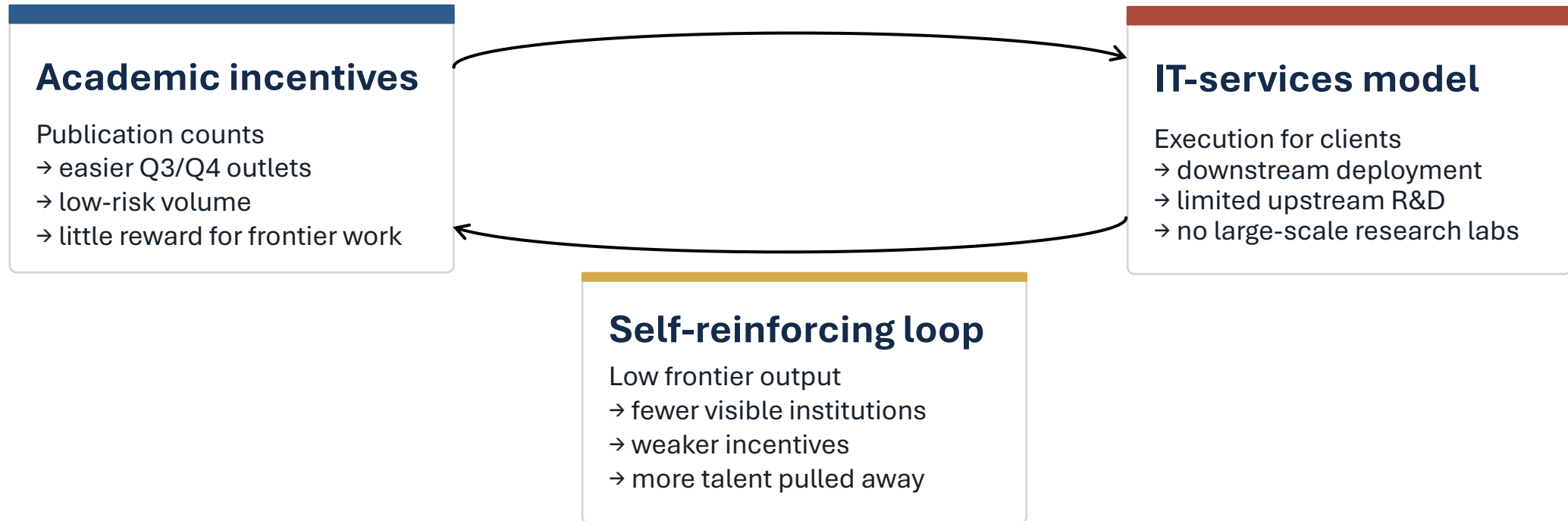
Beyond GPUs, funding, and data centres, India's next AI need is far simpler: a research paper the rest of the world cannot ignore.



Mohit Pandey
Journalist

AUGUST 4, 2026, 4:00 PM

Diagnosis #2 and #3: Misaligned execution



The binding constraint is talent density

1.4%

India's share of papers in top 10 AI conferences, 2018–23

70%

of India's top-conference AI contribution from 8 institutes

90%

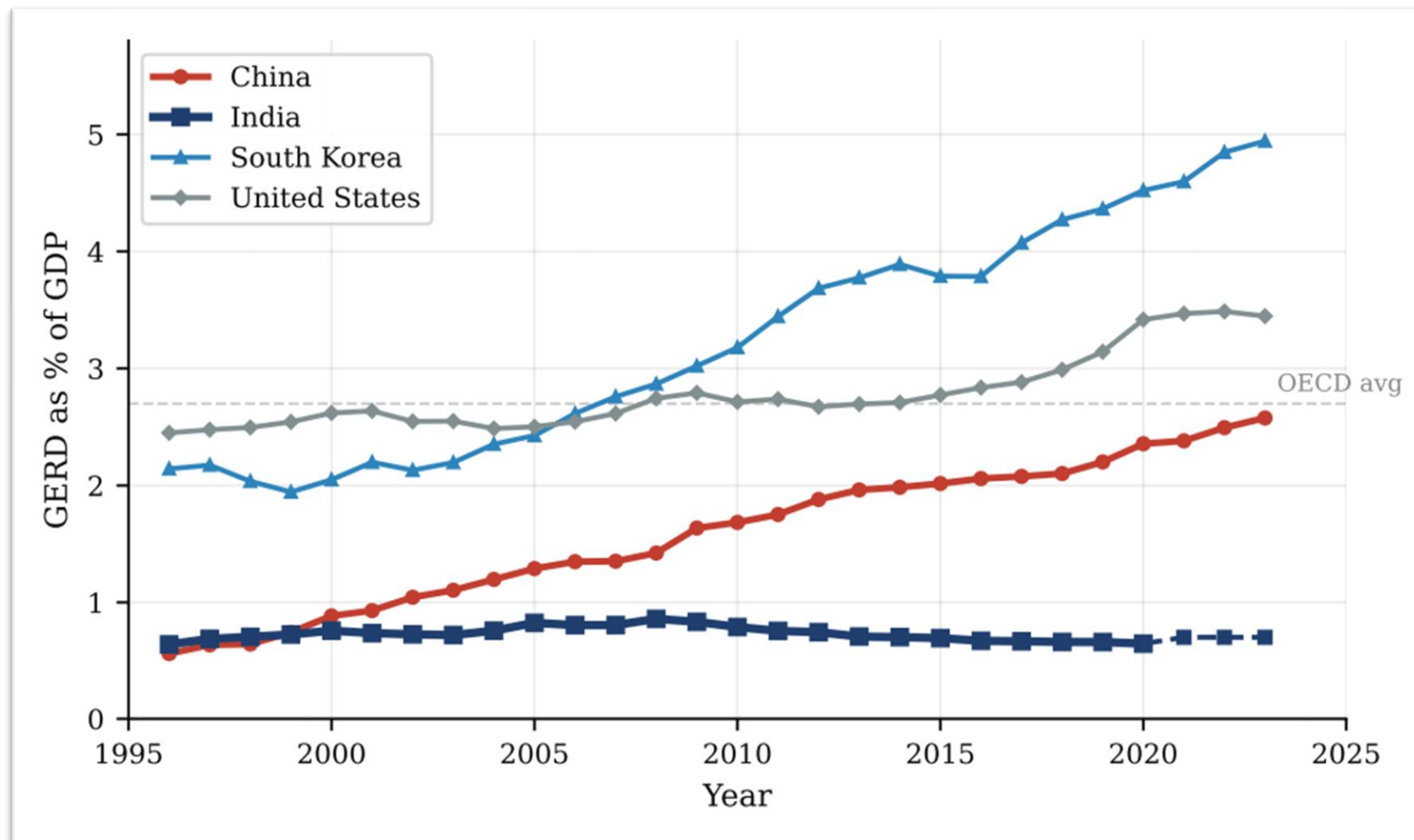
of India's AI papers from 20 institutions

This concentration is an opportunity and a warning.

Source: Change Engine, "India's State of AI Research" (2018–2023)

Counterfactuals: China and South Korea

GERD =
Gross
expenditure
on R&D



The DeepSeek success story

- Little known name in the Chinese tech sector
- Operated under US chip export controls that restricted access to advanced GPUs
- Produced **DeepSeek R1**, an open-source reasoning LLM that matched frontier American proprietary systems
- **Two innovations enabled the need for advanced GPUs**
 - An efficient neural network architecture
 - A novel reinforcement learning algorithm



Moral of the story: Innovation => Sovereignty

The solution

PUBLIC BACKBONE

- Anusandhan National Research Foundation (ANRF) as the patient capital layer
- Dedicated AI research fund + fund-of-funds
- Transparent international peer review

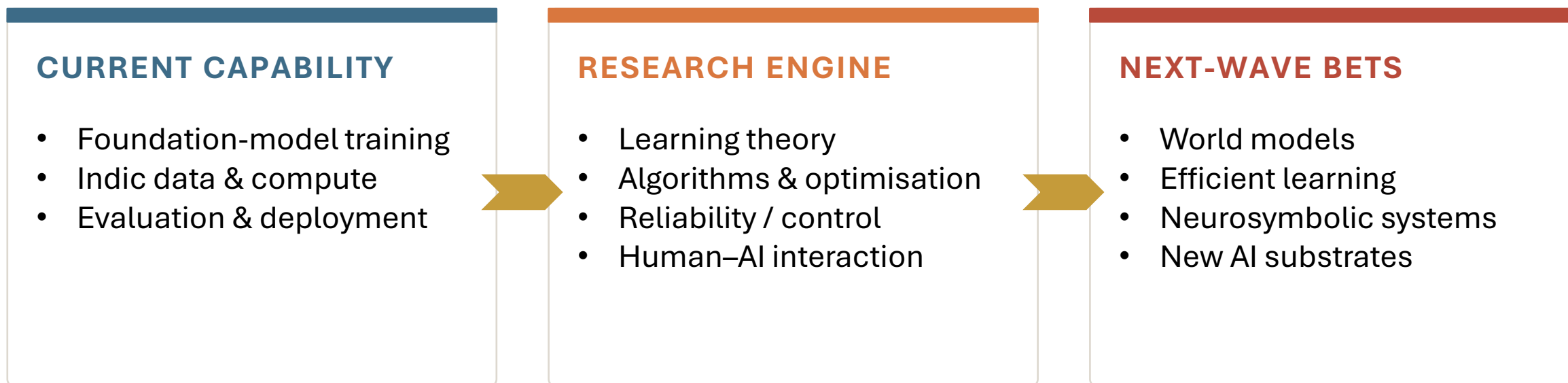
AUTONOMOUS LABS

- Privately or jointly funded frontier research organisations
- Protected from routine government salary / procurement rules
- Long-horizon mandates

UNIVERSITIES + INDUSTRY

- Top-20 research hubs
- Joint / industrial PhDs
- Corporate R&D + research-led product design

Proposal: catch up but prepare for the future



The strategic portfolio should fund all three layers simultaneously.

Five policy moves that are feasible TODAY

- 1 ANRF: dedicated frontier-AI research window**

Fund investigator-led work with top-tier international peer review; patient and application-light.
- 2 Create a handful of autonomous AI labs**

Use CSR + philanthropy + public capital; give salary, hiring and research autonomy.
- 3 Target the top research hubs**

Concentrate resources on the top ~20 institutions; expand faculty, PhDs, postdocs.
- 4 Reform incentives and remove friction**

Reward venue quality/impact; fix procurement and administrative barriers to high-risk research.
- 5 Build global talent pipelines**

Joint PhDs, industrial PhDs, visiting/repatriation programmes, global collaborations.

What success should look like in 5 years

≥5%

Share of global
top-conference
AI contribution

5x

Output at the top
20 Indian AI
institutions

2–3

Frontier research
areas with
sustained
national bets

5–10

World-class
research
institutes with
autonomous
structures



PhD + postdoc
pipeline and
industrial/joint
PhDs

Long-term goal: a research ecosystem that can repeatedly produce frontier capability.

The SwadeshAI vision

Sovereign AI for us does not mean not self-sufficiency based on borrowed tech.

It is strategic agency based on building the future-proof capacity for indigenous innovation.

Food for thought

- Which 2–3 emerging AI directions should India bet on?
- How much autonomy should frontier research institutions receive?
- What can philanthropy and PPPs do before structural R&D reform arrives?

Thank you

Full paper



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